CoderV

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PROBLEM

1. Learning programming is often abstract and difficult to follow.
2. Beginners struggle to understand how code executes step by step.
3. Traditional learning relies heavily on static text and examples.

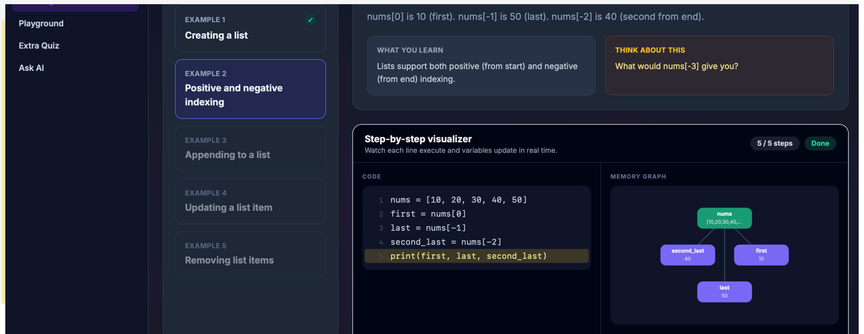
OUR APPROACH

CoderV is an interactive learning platform that makes code execution visual and intuitive.

It allows users to:

1. Observe Python execution step by step.
2. Understand variable changes in real time.
3. Learn using structured lesson and examples
4. Reinforce learning with quizzes.

The system transforms abstract code into a visual learning experience.



SYSTEM OVERVIEW

CoderV is a full-stack educational platform combining visualization, learning content, and assessment tools.

It consists of:

- Frontend (React-based UI)
- Backend (Node.js API server)
- Python execution tracer (sys.settrace)
- Supabase database (storage)
- Firebase authentication (user identity)

VISUALIZATION ENGINE

User code is executed using a Python tracing system that captures:

- Line by line execution
- Variable state changes
- Function calls and returns

This data is sent to the frontend where it is converted into:

- Step-by-step execution view
- Variable tracker
- Graph-based visualization of logic flow

TECH STACK

- Frontend: React, Vite, Tailwind CSS
- Backend: Node.js, Express
- Execution Engine: Python (sys.settrace)
- Database: Supabase (PostgreSQL)
- Authentication: Firebase Auth
- Hosting: Vercel (frontend), Render (backend)

KEY FEATURES

- Step by step Python code visualization
- Interactive node-based execution graph
- Structured lessons (80 examples total)
- Built-in quizzes and assessments
- Progress tracking per user
- Dark/light mode UI
- Free coding playground
- AI Bot to ask about questions

HOW IT WORKS

1. User selects or writes Python code
2. Backend executes code using Python tracer
3. Execution steps are captured as structured data
4. Frontend visualizes:

- Code execution flow
- Variable changes
- Graph relationships

IMPACT & FUTURE WORK

CoderV improves programming understanding by making execution visible and interactive.

Future improvements:

- Advanced graph and tree visualizations
- Mobile friendly interface